

Important Theorems Compendium

BMATS101 – Mathematics-I, III Semester CSE

Statement, proof/justification, and exam usage for every major named theorem across all 5 modules

Module-Wise Theorem Map

- **Module 1 (Curves & Curvature):** No general named theorem – these are specific derivations (pedal equation, radius of curvature), covered in the formula sheet instead.
- **Module 2 (Calculus):** **Euler’s Theorem** on homogeneous functions.
- **Module 3 (ODEs):** No general named theorem – Bernoulli/Clairaut/Lagrange are equation-solving methods, not theorems.
- **Module 4 (Number Theory):** **Fermat’s Little Theorem**, **Wilson’s Theorem**, **Chinese Remainder Theorem**.
- **Module 5 (Linear Algebra):** **Rouché–Capelli Theorem** (consistency of linear systems).
- **Supporting/General:** **Bayes’ Theorem**, **Initial & Final Value Theorems** (Z-transforms), **Taylor’s Theorem** (underlies Maclaurin series).

1 Euler’s Theorem on Homogeneous Functions

Module 2

If $u = f(x, y)$ is a homogeneous function of degree n in x and y (i.e. $f(tx, ty) = t^n f(x, y)$ for all t), then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

More generally, for a homogeneous function of degree n in k variables $x_1, x_2, \dots, x_k$:

$$x_1 \frac{\partial u}{\partial x_1} + x_2 \frac{\partial u}{\partial x_2} + \dots + x_k \frac{\partial u}{\partial x_k} = nu$$

Proof sketch. Since $f(tx, ty) = t^n f(x, y)$, differentiate both sides with respect to t :

$$x f_x(tx, ty) + y f_y(tx, ty) = nt^{n-1} f(x, y)$$

Now put $t = 1$:

$$x f_x(x, y) + y f_y(x, y) = n f(x, y)$$

which is the required result. ■

Extension (Second-order / Euler’s theorem for second derivatives):

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

Where & when it’s used: Whenever a question gives u as a homogeneous function (check: does every term have the same total degree, or can you write u as $x^n \times$ some function of y/x ?) and asks you to “prove that $xu_x + yu_y = nu$ ” or a similar relation. Very common with functions like $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$ or $u = \log\left(\frac{x^3+y^3}{x+y}\right)$ – first find the degree n (careful: for $\sin^{-1}$ or $\log$ of a homogeneous expression, you often apply Euler’s theorem to the *argument* of the function, not u directly, and there’s a modified formula for that case, so check whether the question’s u itself is homogeneous or only becomes homogeneous after removing the outer function).

Modified Euler’s Theorem (for $u = f(v)$ where v is homogeneous)

Statement

If $u = f(v)$ where $v(x, y)$ is homogeneous of degree n , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n v f'(v) = n \frac{v f'(v)}{1}$$

Often rewritten using $g(v) = \frac{f(v)}{f'(v)}$ so that

$$x u_x + y u_y = n g(v), \quad \text{where } g(v) = \frac{f(v)}{f'(v)}$$

Where & when it's used: Use this version specifically when $u = \sin^{-1}(\cdot)$, $u = \tan^{-1}(\cdot)$, $u = \log(\cdot)$, or similar *transcendental* function is applied on top of a homogeneous expression v – direct Euler's theorem does not apply to u itself since u is not homogeneous, only v is.

2 Fermat's Little Theorem

Module 4

If p is a prime number and a is any integer with $\gcd(a, p) = 1$, then

$$a^{p-1} \equiv 1 \pmod{p}$$

Equivalently (without the coprimality restriction): $a^p \equiv a \pmod{p}$ for every integer a .

Proof sketch. Consider the set $\{a, 2a, 3a, \dots, (p-1)a\} \pmod{p}$. Since $\gcd(a, p) = 1$, these are a permutation of $\{1, 2, \dots, p-1\} \pmod{p}$. Multiplying all elements:

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$$

Since $\gcd((p-1)!, p) = 1$, we can cancel $(p-1)!$ from both sides, giving $a^{p-1} \equiv 1 \pmod{p}$. ■

Where & when it's used: This is the single most useful tool in Module 4. Use it to: (1) reduce a huge exponent modulo a prime – e.g. find $2^{1000} \pmod{13}$ by reducing $1000 \pmod{13-1} = 12$ first; (2) find last digits of large powers combined with cycle detection; (3) as a stepping stone to RSA encryption/decryption correctness proofs. Trigger phrase in questions: "find the remainder when a^n is divided by a prime p ."

3 Wilson's Theorem

Module 4

If p is a prime number, then

$$(p-1)! \equiv -1 \pmod{p}$$

Conversely, if $(n-1)! \equiv -1 \pmod{n}$, then n is prime (this converse is sometimes used as a primality test).

Proof sketch. Pair up each element $a \in \{1, \dots, p-2\}$ (excluding 1 and $p-1$) with its multiplicative inverse mod p . Every such element has a unique inverse in the same range, and $a = a^{-1}$ only when $a \equiv \pm 1 \pmod{p}$. So all elements from 2 to $p-2$ pair up into products $\equiv 1 \pmod{p}$, leaving $(p-1)! \equiv 1 \cdot (p-1) \equiv -1 \pmod{p}$. ■

Where & when it's used: Use whenever a question asks for a factorial modulo a prime, e.g. "find the remainder when $14!$ is divided by 17 ." Since 17 is prime, $16! \equiv -1 \pmod{17}$ by Wilson's theorem; then work backward by dividing out 15×16 (using modular inverses) to get $14! \pmod{17}$.

4 Chinese Remainder Theorem (CRT)

Module 4

Let $m_1, m_2, \dots, m_k$ be pairwise coprime positive integers, and let $M = m_1 m_2 \cdots m_k$. Then the system of simultaneous congruences

$$x \equiv a_1 \pmod{m_1}, \quad x \equiv a_2 \pmod{m_2}, \quad \dots, \quad x \equiv a_k \pmod{m_k}$$

has a *unique* solution modulo M , given by

$$x \equiv \sum_{i=1}^k a_i M_i y_i \pmod{M}, \quad \text{where } M_i = \frac{M}{m_i}, \quad y_i \equiv M_i^{-1} \pmod{m_i}$$

Proof idea. Existence: construct $x = \sum a_i M_i y_i$ and verify it satisfies each congruence individually (since $M_i \equiv 0 \pmod{m_j}$ for $j \neq i$, only the i -th term survives mod m_i , and $M_i y_i \equiv 1 \pmod{m_i}$ by construction). Uniqueness: if x_1, x_2 both satisfy all congruences, then $m_i \mid (x_1 - x_2)$ for every i , so (by pairwise coprimality) $M \mid (x_1 - x_2)$. ■

Where & when it's used: Use whenever a question gives two or more simultaneous congruences with different (coprime) moduli and asks for the smallest positive x satisfying all of them – a very standard Module 4 question type. Steps: find M , then M_i for each modulus, then the inverse y_i of $M_i \pmod{m_i}$ (often by trial for small moduli), then combine.

5 Rouché–Capelli Theorem (Consistency of Linear Equations)

Module 5

A system of linear equations $Ax = b$ (with A the coefficient matrix, $[A|b]$ the augmented matrix) is consistent if and only if

$$\text{rank}(A) = \text{rank}([A|b])$$

Furthermore, if consistent and this common rank equals the number of unknowns n , the solution is *unique*; if the common rank is less than n , there are *infinitely many* solutions (with $n - \text{rank}$ free parameters). If $\text{rank}(A) \neq \text{rank}([A|b])$, the system has *no solution*.

Justification (sketch). Row-reducing $[A|b]$ to echelon form: if a row appears as $[0 \ 0 \ \cdots \ 0 \ | \ c]$ with $c \neq 0$, this represents the equation $0 = c$ (impossible), which happens exactly when the augmented matrix has strictly higher rank than A alone – hence inconsistency. Otherwise every equation reduces to solvable form, giving either a unique solution (full rank) or infinitely many (rank deficient, free variables remain). ■

Where & when it's used: Use this immediately whenever a question says "test the consistency of the following system and solve if consistent." Procedure: reduce $[A|b]$ to row echelon form, compare $\text{rank}(A)$ with $\text{rank}([A|b])$, then classify using the three cases above before attempting to actually solve.

6 Bayes' Theorem

Probability / general reference

Let $B_1, B_2, \dots, B_n$ be mutually exclusive and exhaustive events (a partition of the sample space), and let A be any event with $P(A) > 0$. Then for each i :

$$P(B_i | A) = \frac{P(B_i) P(A | B_i)}{\sum_{j=1}^n P(B_j) P(A | B_j)}$$

Proof sketch. By definition of conditional probability, $P(B_i | A) = \frac{P(B_i \cap A)}{P(A)}$ and $P(A | B_i) = \frac{P(B_i \cap A)}{P(B_i)}$, so $P(B_i \cap A) = P(B_i) P(A | B_i)$. Also, since the B_j partition the sample space, $P(A) = \sum_j P(B_j \cap A) = \sum_j P(B_j) P(A | B_j)$ (law of total probability). Substituting gives the result. ■

Where & when it's used: Use whenever a question describes several possible "causes" or "sources" (e.g. items from different machines/factories) with known prior probabilities, gives the probability of an "effect" (e.g. defective item) under each cause, and then asks for the probability of a particular cause *given* that the effect was observed – this reversal of conditioning is the signature of a Bayes' theorem question.

7 Initial Value Theorem & Final Value Theorem (Z-Transforms)

Z-transforms / general reference

Initial Value Theorem: If $Z(u_n) = U(z)$, then

$$u_0 = \lim_{z \rightarrow \infty} U(z)$$

Final Value Theorem: If $Z(u_n) = U(z)$ and $\lim_{n \rightarrow \infty} u_n$ exists, then

$$\lim_{n \rightarrow \infty} u_n = \lim_{z \rightarrow 1} (z - 1)U(z)$$

Justification (sketch). From the definition $U(z) = \sum_{n=0}^{\infty} u_n z^{-n} = u_0 + u_1 z^{-1} + u_2 z^{-2} + \dots$. As $z \rightarrow \infty$, every term with $z^{-1}, z^{-2}, \dots$ vanishes, leaving $U(z) \rightarrow u_0$. The final value result follows from applying the shifting property to $(z-1)U(z)$ and taking the limit as $z \rightarrow 1$, which telescopes the partial sums of u_n down to $\lim_{n \rightarrow \infty} u_n$. ■

Where & when it's used: Use these to extract the first term (u_0) or the long-run/steady-state value of a sequence directly from its Z-transform $U(z)$, without having to compute the full inverse Z-transform first. Very useful as a quick check or a shortcut in exam problems that ask specifically for u_0 or u_{∞} .

8 Taylor's Theorem

Underlies Module 2 series expansion

If $f(x)$ is a function that is $(n+1)$ times differentiable near $x = a$, then for x near a :

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots + \frac{(x-a)^n}{n!} f^{(n)}(a) + R_n$$

where R_n is a remainder term (e.g. in Lagrange form, $R_n = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(\xi)$ for some ξ between a and x).

When $a = 0$, this is called **Maclaurin's Theorem/Series**.

Why it works (intuition). Each successive term is chosen so that the partial sum matches f and all its derivatives up to order n exactly at $x = a$; the remainder term R_n captures the error from truncating the (potentially infinite) series, and shrinks as more terms are added provided the function is well-behaved (analytic) near a .

Where & when it's used: This is the theoretical justification behind every Maclaurin/Taylor series expansion question in Module 2 (e.g. expanding $\sqrt{1+\sin 2x}$, $\log(\sec x)$). You won't be asked to prove Taylor's theorem itself in BMATS101, but understanding it explains *why* the term-by-term differentiation method used in Maclaurin expansions is valid.

Compiled for BMATS101 (Mathematics-I, CSE Stream) based on the modules covered in the VTU exam papers and Model Question Papers, cross-checked against the standard III-semester Mathematics Handbook. Theorems are grouped by the module in which they are actually examined.